Answer Memory and Reader Memory in Random 3-SAT

Exact residual widths, reader carry and construction cost under sealed prediction

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Abstract

A Boolean function read in a fixed variable order has, at every cut, an exact minimal memory: the number of residual functions that the unread variables can still tell apart. This paper separates three memories that are usually treated as one. The answer's memory is that residual quotient. The reader's memory is what a procedure that reads the clauses must carry between steps. The construction memory is what a compiler must hold on its way to the answer's quotient. Six experiments on random 3-SAT, each sealed by SHA-256 before its data was linked, and two structured families place the cost. At 4.26 clauses per variable the answer's memory is about two bits and does not grow with n: its fitted slope is 0.02 per variable over n = 10 to 20. Every reader grows exponentially. The clause-status reader grows as $2^{0.43n}$ in fixed orders, and as $2^{0.22n}$ on the best possible route for each instance, found exactly over all prefix sets (0.28 per variable over the last three sizes). The reduced decision diagram built clause by clause passes through $2^{0.29n}$ nodes while its final size grows as $2^{0.11n}$. A Davis–Putnam resolution kernel holds $2^{0.50n}$ live clauses. A sweep over clause density at n = 18 shows the inversion directly: as density rises, the answer's memory falls from $2^{6.8}$ states to a single state, the construction peak stays between $2^{7.2}$ and $2^{9.4}$, and the gap between them is largest past the threshold, where the answer is empty. The best possible crossing boundary is loose: the best reader uses about a tenth of its exponent. Two structured families show the mechanism. Mirrored equalities fall from $2^{n/2}$ states to 3 when read in the ring's mirror order, and four-clause blocks carry 4^m clause statuses for one residual state. The answer is small exactly where the reader is large. What remains open is a reader that closes a constraint web that crosses everywhere without enumerating its crossings.

> **Status words. STANDS**: proved here, or established in the literature. **RUNS**: an exact computation landed where a prediction sealed before the run placed it; the range is stated. **measured**: an exact computation reported without a sealed prediction. **ELSEWHERE**: the computation ran and landed, and the landing belongs to a different owner, which is named. **OPEN**: stated, with its missing link named. **PULL**: a direction the results point to, not yet routed.

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1. Introduction

The cost of a hard problem has to live somewhere. It can live in the answer, as the size of the shape the solutions make. It can live in the reader, as what a procedure must hold between steps. Or it can live in the construction, as what a compiler must pass through between the question's short description and the answer's shape. These three places are usually treated as one, because the usual measure of difficulty, running time, adds them together. This paper measures them separately, on the same formulas, under predictions sealed before the data.

The program this work belongs to holds that a reader is never inert. What a reader retains is part of the transformation it performs, and the cost of a transformation is paid in the reader, not in the field it reads. The cut-law series made that exact for the adder: at every cut of an n-position adder, the reader must retain exactly the classes of the past that the future can still distinguish [13, 14, 15]. The same path at a different location is random 3-SAT, where the field is a formula and the classes are residual Boolean functions.

The answer's memory is a classical object. Fix an order of the variables and a cut after k of them. Every assignment to the first k variables leaves a residual function of the rest, and the number of distinct residuals is the minimal number of states any deterministic reader in that order must distinguish. This is the Myhill–Nerode quotient of the function at that cut [3], and it equals the width at that level of the reduced ordered binary decision diagram [1, 2].

Random 3-SAT has a well-mapped landscape. Formulas with about 4.27 clauses per variable sit at the satisfiability threshold predicted by the cavity method [12]. Search procedures are hardest near it [9, 10]. The location of the fifty-percent point shifts with size [11]. Resolution, the proof system behind clause learning, requires exponential size on random unsatisfiable 3-CNF at densities its lower-bound theorems cover [5, 6, 7, 8]. These results serve here as comparators. None of them separates the answer's memory from the reader's.

The paper makes six contributions.

- Exact relations between the three memories: the answer's width is at most the number of solutions plus one; the clause-status carry lies between the answer's width and 2 raised to the number of crossing clauses, plus one; no fixed order is privileged on uniform random 3-SAT; two structured families have exact width laws (Section 2). **STANDS.**
- Six sealed experiments that measure the three memories on the same distribution, from fixed orders to the best possible route per instance, a compiler that builds the answer's shape, and a resolution kernel (Sections 4 to 9).
- The inversion: at the hard density the answer's memory is about two bits, flat in n, while every reader grows exponentially, and across densities the answer shrinks toward the threshold while the construction cost stays (Sections 4 and 8).
- A measurement of how loose the syntactic boundary is: the best possible order still crosses about 2.4 clauses per variable, and the best reader uses about a tenth of that exponent (Section 9).

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- The ring's place: its mirror order is privileged exactly when a constraint web follows mirror pairs, and placing an arbitrary web onto mirror pairs is the variable-ordering problem in ring coordinates (Section 10).
- A complete ledger of every sealed prediction and where it landed (Section 13).

2. Three memories

2.1 Cuts and residuals

Let F be a Boolean function of n variables and π an order of the variables. A cut after k variables splits an assignment into a prefix a, the values of the first k variables in π, and a suffix b, the values of the rest. The residual of a prefix is the function $F_a(b) = F(a, b)$. The **answer's width** at the cut is

$$W_k = |\{ F_a : a \in \{0, 1\}^k \}|.$$

Two prefixes with the same residual can share one reader state, and two with different residuals cannot, because some suffix separates them. So W_k is the minimal number of states any deterministic reader in order π must distinguish at cut k, and the reduced ordered decision diagram of F in order π has exactly the nonconstant residuals as its nodes at that level [1, 2, 3]. The profile $W_0, \ldots, W_n$ is the answer's memory. It is a shape: the past quotiented by everything the future cannot tell apart.

2.2 The value reader

A procedure that reads the clauses of a CNF formula carries a concrete state at each cut. If some clause whose variables have all been read is already false, the prefix is dead. Otherwise the state records which of the clauses crossing the cut, those with variables on both sides, are already satisfied. The **value-reader carry** C_k is the number of distinct states that prefixes realize at the cut. This reader reads the clauses and the values written so far: it carries what has happened. The residual of an alive prefix is determined by its state, so the value reader is correct, and it is generally not minimal.

2.3 The construction memory

A compiler can build the answer's shape from the formula without ever expanding the truth table. Two such compilers are measured here. The first conjoins the clauses one at a time into a reduced ordered decision diagram, merging equal futures as it goes; its construction memory is the peak number of live nodes over the stages [1]. The second is Davis–Putnam elimination, which resolves variables out one at a time [4]; its construction memory is the peak number of live clauses. A third reader, DPLL search with unit propagation, is the classical yardstick of hardness and is reported by its number of search nodes.

2.4 Exact relations

Theorem 1 (the answer is bounded by its solutions). For every order and every cut, $W_k \le s + 1$, where s is the number of satisfying assignments of F.

Proof. A prefix whose residual is not the constant 0 extends to at least one satisfying assignment, and distinct prefixes extend to disjoint sets of assignments. So at most s prefixes have nonzero residuals, there are at most s distinct nonzero residuals, and the zero residual adds at most one more. ■

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So an unsatisfiable formula has width 1 at every cut, and a formula with one solution has width at most 2. This bound appears in the independent audit of this work.

Theorem 2 (the reader lies between the answer and the boundary). Let b_k be the number of clauses crossing cut k. Then $W_k \le C_k \le 2^{b_k} + 1$.

Proof. Every dead prefix has residual 0. The residual of an alive prefix is the conjunction of the clauses on the unread side and of the crossing clauses not yet satisfied, restricted to the unread variables, so it is a function of the reader's state. That gives $W_k \le C_k$. An alive state is a subset of the b_k crossing clauses, and the dead state adds one. ■

Theorem 3 (no fixed order is privileged on uniform random 3-SAT). Under the generator of Section 3, every fixed variable order, and every random order chosen independently of the formula, gives the same distribution of width profiles W and carry profiles C.

Proof. The generator's distribution is invariant under every permutation of the variable names. Reading a formula in order σ is reading the relabeled formula in the identity order, and the relabeled formula has the same distribution. ■

The theorem was stated in the independent audit before its runs. It means that the ring's mirror order can have no systematic advantage on this distribution, and any difference between fixed orders in a finite sample is sampling noise.

Theorem 4 (mirrored equalities). For even n let $E_n = \bigwedge_{i<n/2} (x_i \leftrightarrow x_{n-1-i})$, written as 2-CNF. In the binary order the width at the midpoint is $2^{n/2}$. In the ring's outside-in mirror order 0, n−1, 1, n−2, … the width never exceeds 3 for $n \ge 4$.

Proof. In the binary order, after the first n/2 variables every bit is an unresolved obligation on a distinct unread mirror bit, and each of the $2^{n/2}$ prefixes leaves a distinct residual. In the mirror order each equality closes before the next begins, so at any cut at most one equality is open, and its residual is one of three functions: false, the requirement that the partner bit equal 0, or that it equal 1. ■

Theorem 5 (four-clause blocks). Let $F_m = \bigwedge_{i\le m} \bigwedge (x_i \vee \pm y_i \vee \pm z_i)$ over all four sign patterns. Then $F_m \equiv \bigwedge x_i$. In an order that reads every y_i and z_i before any x_i, at the cut after the last of them the value-reader carry is 4^m and the answer's width is 1.

Proof. Resolving the two z-opposite pairs and then the y-opposite pair gives x_i, so $F_m \equiv \bigwedge x_i$. A pair of values for (y_i, z_i) falsifies the y and z literals of exactly one of the block's four clauses, so the four value pairs leave four different sets of satisfied clauses, and the blocks combine independently into 4^m states. Every one of those prefixes leaves the same residual, $\bigwedge x_i$. ■

Theorem 6 (placement is ordering). Write M_n for the ring's mirror order. Every order π is M_n composed with a relabeling f, $f(M_n(k)) = \pi(k)$. So placing a constraint web onto the ring's mirror pairs to minimize memory is the variable-ordering problem written in ring coordinates. The coordinate change is free once f is known; finding the best f is not.

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Theorems 4, 5 and 6 were derived in the independent audit; Section 10 reproduces Theorems 4 and 5 by exhaustive computation. **STANDS** for all six.

3. Protocol

Generator. A formula on n variables has m = round(αn) clauses. Each clause draws three distinct variables uniformly and gives each literal an independent random sign; clauses may repeat. Every program uses a fixed seed, so every table below is reproduced exactly by the commands in Appendix B.

Orders. The binary order reads variable 0 first. The mirror order is the ring's outside-in order 0, n−1, 1, n−2, … [16]. A random order is drawn per formula. Two greedy orders are used: in Section 4 the next variable leaves the fewest crossing clauses, and in Section 5 it leaves the fewest value-reader states. The best route is found exactly. Both W and C at a cut depend only on the set of variables read, not on the order in which they were read, so the best possible peak over all orders is a bottleneck path through the 2^n subsets, computed by dynamic programming.

Measurement. W is computed from the full truth table, as the number of distinct rows of the $2^k \times 2^{n-k}$ truth matrix. This is a measurement method, exponential by design, and no reader uses it. C is computed by enumerating prefixes. The decision diagram is a canonical reduced ordered diagram without complemented edges. Its clauses are scheduled by the position of their last variable, and its live size is counted after each clause. The resolution kernel always eliminates the variable whose elimination adds the fewest clauses, drops tautologies and merges duplicates. DPLL branches on the most frequent variable, true first, after unit propagation.

Sealing. Each experiment's predictions were written as text and hashed with SHA-256 before the program that tests them ran. The six digests are in Appendix A, and the companion notebook verifies them. Medians are over satisfiable formulas unless stated. Slopes are least-squares fits of the $\log_2$ medians against n over the stated range. They are finite-range readouts, not asymptotic laws.

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4. Experiment 1: the answer and the value reader in fixed orders

n	SAT	answer's memory: binary, mirror, random, greedy	value reader: binary, mirror, random, greedy
10	12/16	1.58, 1.58, 1.58, 1.58	4.86, 5.02, 4.67, 4.39
12	13/16	2.00, 2.00, 1.58, 1.58	5.49, 5.49, 5.70, 5.21
14	11/16	2.00, 2.00, 2.00, 2.00	6.52, 6.67, 6.79, 5.98
16	10/16	2.16, 2.16, 2.16, 2.16	7.60, 7.49, 7.32, 7.18
18	12/16	1.79, 1.58, 1.58, 1.79	8.35, 8.35, 7.97, 7.78
20	11/16	2.00, 2.00, 2.00, 2.00	9.01, 9.39, 9.18, 8.29
slope	—	0.023, 0.014, 0.032, 0.041	0.435, 0.446, 0.427, 0.406

Table 1. Peak memory, median $\log_2$ over satisfiable formulas, 4.26 clauses per variable, 16 formulas per size. The median number of solutions runs from $2^{1.6}$ to $2^{3.5}$.

At the hard density the answer's memory is between three and five states at every size, and its fitted growth is 0.01 to 0.04 per variable. The value reader grows at 0.41 to 0.45 per variable in every order. The four orders agree within sampling noise, as Theorem 3 requires; the ring's mirror order has no advantage here because this distribution has no mirror structure to follow. Every unsatisfiable formula has width 1 at every cut.

The same measurement in the easy region, at 2.0 clauses per variable with 10 formulas per size from n = 10 to 18, turns the answer around. There formulas have thousands of solutions (a median of $2^{10.6}$ at n = 18), and the answer's memory grows at 0.34 per variable. The value reader grows at 0.39 to 0.59. **The answer's memory is large where problems are easy and small where they are hard.**

Ledger. **P1 RUNS**: fixed orders give equal width distributions. **P2 ELSEWHERE**: the sealed prediction placed exponential growth in the answer's memory; the answer's memory is flat, and the growth belongs to the reader's representation. **P3 ELSEWHERE**, for the same owner: the greedy order cannot lower a width that does not grow. **P4 RUNS**: unsatisfiable formulas have width 1. The value-reader column was added after the first readout and is **measured**.

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5. Experiment 2: the best route for each instance

n	binary	mirror	greedy by carry	best route over all orders
8	3.95	3.91	3.46	3.00
10	4.86	4.86	3.64	3.08
12	5.98	6.02	4.61	3.58
14	6.84	6.44	5.11	4.17
16	7.50	7.56	5.49	4.70
slope	0.454	0.444	0.277	0.224

Table 2. Peak value-reader carry, median $\log_2$ over satisfiable formulas at 4.26 clauses per variable; 12 formulas per size, 8 at n = 16. The best route is exact: a bottleneck path through all 2^n prefix sets.

Letting each formula choose its own route cuts the exponent roughly in half. The best route lies strictly below every fixed order on 55 of 56 formulas and ties on the last. Its growth is 0.22 per variable over the whole range and 0.28 over the last three sizes, with no sign of bending toward zero. The best route is a property of the instance, and it is found by visiting every prefix set, so it prices the reader's representation, not a procedure that a reader could run cheaply (Technical note B).

Ledger. **Q1 RUNS**: the best route lies below every fixed order. **Q2 RUNS**: its peak still grows linearly in $\log_2$, above 0.15 per variable. **Q3 RUNS**: the mirror order is one fixed order among others on this distribution.

6. Experiment 3: the shape reader built from the formula

n	final size	peak on the way	nodes ever made	solutions	peak, unsatisfiable
10	3.91	5.23	8.25	1.79	4.91
14	4.46	6.39	9.48	2.00	5.86
18	5.09	7.61	10.95	2.16	7.53
22	5.05	8.63	12.03	2.66	8.18
26	5.73	10.12	13.50	4.11	10.08
30	6.00	11.10	14.56	4.00	10.81
34	7.41	12.49	16.34	6.13	12.33
38	7.15	13.06	16.81	5.52	12.98
42	7.98	14.31	18.11	7.11	13.25
46	7.59	15.85	19.29	4.88	14.63
slope	0.114	0.289	0.308	—	0.272

Table 3. The reduced ordered decision diagram conjoined one clause at a time, 4.26 clauses per variable, 12 formulas per size. Median $\log_2$ node counts; the first four columns are over satisfiable formulas, the last over unsatisfiable ones.

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The compiler reads the constraints, not assignment histories, and merges equal futures at every stage, so it is a reader of shape. Its final diagram is small, within the bound of Theorem 1 on every formula. Its intermediate stages are not. The peak grows steadily at $2^{0.29n}$ through n = 46, and the excess of peak over final widens from $2^{1.3}$ at n = 10 to $2^{8.3}$ at n = 46. Unsatisfiable formulas end at the single 0 terminal and still pass through $2^{0.27n}$ nodes on the way.

The independent audit built the same compiler separately, for 96 formulas at n = 10 to 20 in both binary and mirror order, and measured the exact residual width of every intermediate function rather than its node count. At n = 20 the median peak intermediate width was 216.5 states against a final width of 3.5. One satisfiable formula at n = 20 with exactly one solution has a final width of 2, and its intermediate widths peak at 200 in binary order and at 323 in mirror order.

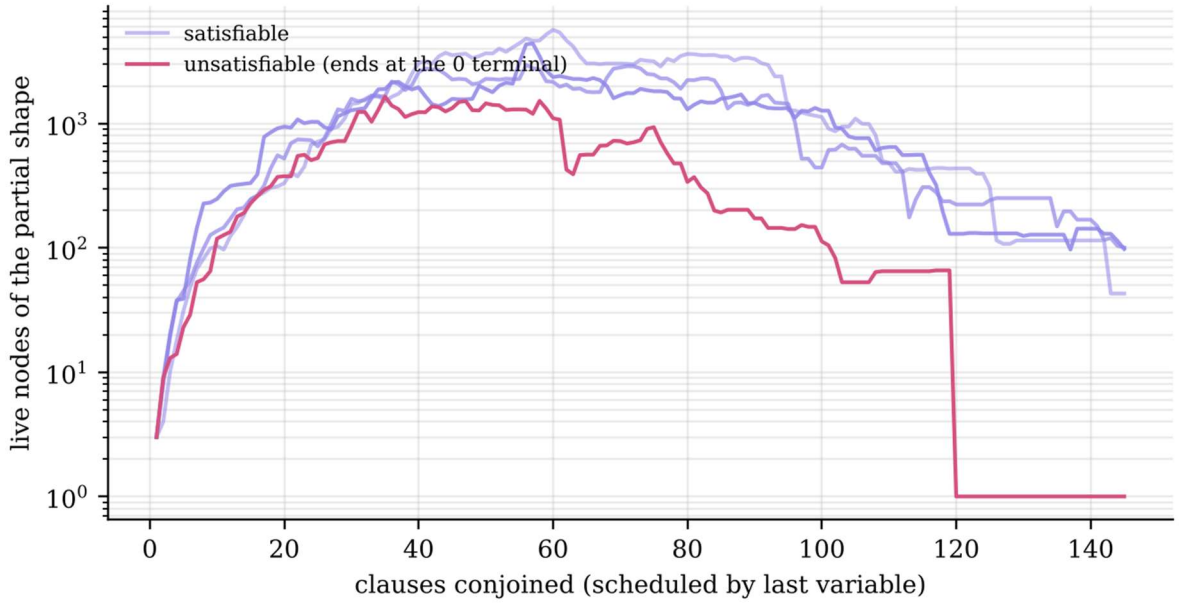


Figure 1. The swell. Live nodes of the partial shape after each clause, for three satisfiable formulas and one unsatisfiable formula at n = 34. Every partial formula has many solutions and therefore a large shape; the answer's shape comes into view only as the last clauses land.

Ledger. **R1 RUNS**: every final shape is within s + 1 nodes per level. **R2 RUNS**: the peak grows at 0.29 per variable. **R3 RUNS**: peak over final grows with n. **R4 RUNS**: unsatisfiable formulas swell before they vanish.

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7. Experiment 4: a resolution kernel

n	peak live clauses	resolvents formed
10	6.49	8.74
12	7.50	10.55
14	8.51	12.05
16	9.57	14.01
18	10.88	16.36
20	11.26	16.89
slope	0.50	0.86

Table 4. Davis–Putnam elimination, fewest added clauses first, 4.26 clauses per variable, 10 formulas per size, all formulas. Median $\log_2$ counts.

Resolution closes a relationship before any value is visited: it removes a variable from every clause that mentions it at once. That is a reader of the leading edge, and it works perfectly where the structure allows. On the four-clause blocks of Theorem 5 the kernel closes every block with no swell at all: its peak is exactly the 4m input clauses, and it forms no resolvents, because each x_i occurs only positively and its block closes as a pure literal. On random 3-SAT the kernel's peak grows at $2^{0.50n}$ and its total work at $2^{0.86n}$. This kernel keeps subsumed clauses, so its exponent reads high for its family (Technical note C). The lower-bound theorems for resolution on random unsatisfiable 3-CNF [5, 6, 7, 8] are the comparator, at the densities they cover. Chvátal and Szemerédi require $c \cdot 2^{-k} \geq 0.7$, which for 3-CNF is at least 5.6 clauses per variable [5].

Ledger. **D1 RUNS**: the peak grows above 0.15 per variable. **D2 RUNS**: every formula with n ≤ 20 is decided correctly against brute force, 60 of 60, and none reaches the 400,000-clause cap; the largest peak is 4,989 clauses. **D3**: the peak of 4m clauses with no swell **RUNS**; the sealed count of three resolutions per block lands **ELSEWHERE**, owned by pure-literal elimination, which closes each block with none.

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8. Experiment 5: every density at one size

α	SAT	answer	reader	build peak	build final	resolution	DPLL	build – answer
1.0	1.00	6.81	8.13	8.97	8.97	4.17	3.00	2.14
1.5	1.00	6.83	9.12	9.36	9.27	4.75	3.17	2.43
2.0	1.00	6.62	9.82	9.32	9.19	5.17	3.00	2.77
2.5	1.00	5.52	9.91	8.90	8.34	5.49	2.81	3.26
3.0	1.00	4.58	9.31	8.38	7.61	6.79	3.00	3.80
3.5	0.96	3.39	9.16	8.01	6.48	8.30	2.90	4.51
4.0	0.88	2.32	8.58	7.81	5.57	9.94	3.17	5.55
4.26	0.79	1.00	8.15	7.56	4.17	10.52	3.80	6.32
4.5	0.71	1.29	8.22	7.61	4.38	10.65	3.70	6.08
5.0	0.58	1.00	7.68	7.16	4.13	11.81	3.38	6.39
5.5	0.08	0.00	7.63	7.19	0.00	12.63	3.70	7.11

Table 5. The density sweep at n = 18, binary order. Medians of $\log_2$ counts: the answer's width, the value-reader carry, the decision diagram's peak and final sizes, the resolution kernel's peak, DPLL search nodes, and the median per-formula gap between the construction peak and the answer's width.

The sweep holds the size fixed and lets the constraint web thicken. The answer's memory falls monotonically from $2^{6.8}$ states to a single state. The value reader rises to $2^{9.9}$ at 2.5 clauses per variable and then declines slowly. The construction peak declines only from $2^{9.4}$ to $2^{7.2}$. The resolution kernel rises throughout, to $2^{12.6}$. The gap between construction and answer grows from 2.1 bits to 7.1 bits. DPLL search, the classical yardstick, peaks at 4.26, the density where the answer has just become small; at this size the peak is shallow (Technical note D). The supplementary run at 6.0 and 7.0 continues the pattern: the answer is the single 0 state, the construction peak is $2^{6.8}$ and $2^{6.6}$, and DPLL falls back to $2^{3.5}$.

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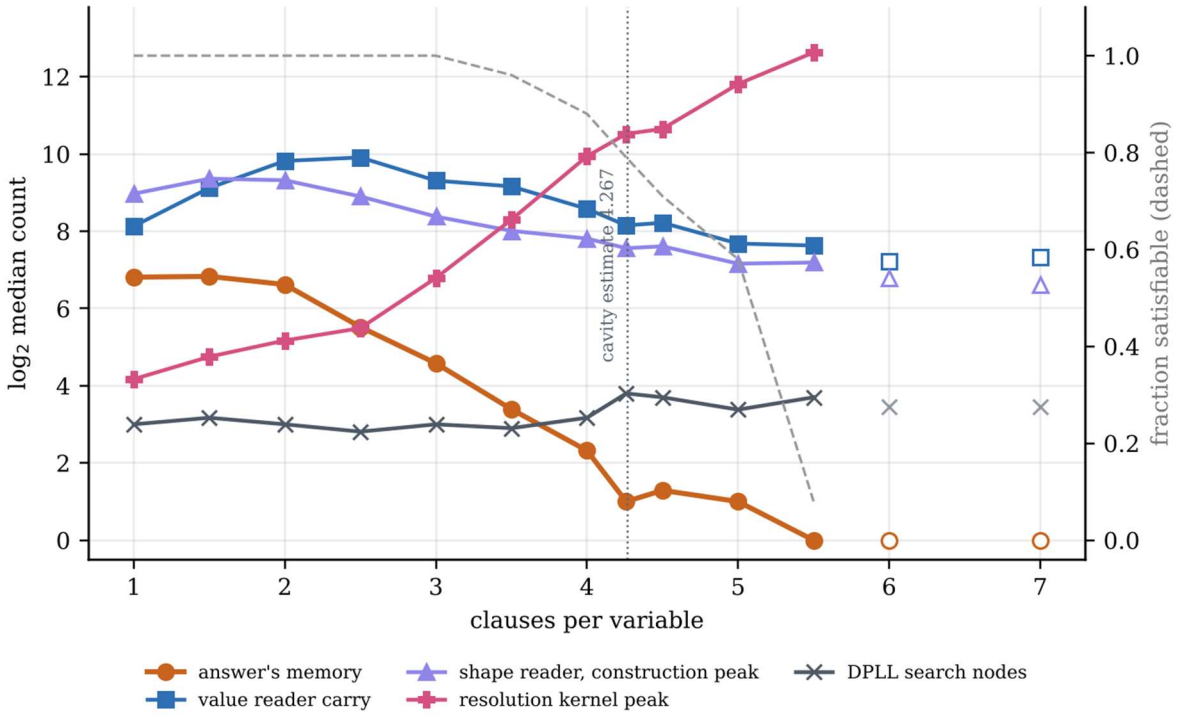


Figure 2. Five memories of the same formulas at n = 18, against clause density; medians of $\log_2$ counts over all formulas, 24 per density. Open markers at 6.0 and 7.0 are a supplementary run without the resolution kernel. The dashed line is the fraction satisfiable (right axis).

Ledger. **E1 RUNS**: the DPLL median peaks within [3.5, 5.0], at 4.26. **E2 RUNS**: the answer's memory is largest at 1.5 clauses per variable and at most 2.45 states from 4.26 on. **E3 ELSEWHERE**: the gap is largest at 5.5, past the threshold. Its owner is the empty answer: once the answer is the single 0 state, the construction still costs about 2^7. **E4 ELSEWHERE**: at n = 18 the satisfiable fraction crosses one half between 5.0 and 5.5. The owner is the size, since the fifty-percent point of small formulas sits above the asymptotic estimate of 4.267 [11, 12]. Densities 6.0 and 7.0 were not completed under the seal, because the resolution kernel's running time grew past the budget; they are reported from the supplementary run as **measured**.

9. Experiment 6: how loose the boundary is

n	best boundary b*	best carry, $\log_2$ c*	$\log_2$ c* / b*, median	$\log_2$ c* / b*, largest
8	24.5	3.17	0.127	0.151
10	30.0	3.32	0.111	0.123
12	34.0	3.64	0.112	0.121
14	39.0	4.04	0.105	0.122

Table 6. The smallest possible peak number of crossing clauses over all orders, b, against the smallest possible peak value-reader carry, c*; both exact by subset dynamic programming. Satisfiable formulas at 4.26 clauses per variable, 12 per size.*

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Even the best possible order leaves 24 to 39 clauses crossing its worst cut, and that boundary grows at 2.4 clauses per variable. Theorem 2 would permit up to 2^{39} reader states at n = 14; the best reader needs $2^{4.0}$. The ratio of exponents is about one tenth at every size and does not rise. Most of the crossing clauses never become distinct futures. The syntactic boundary is a burden that the formula's own algebra mostly closes, and counting crossing clauses overstates memory by a factor of about nine in the exponent.

Ledger. **F1 RUNS**: b∗ grows linearly in n. **F2 RUNS**: every ratio is below one half; the largest is 0.151. **F3 RUNS**: the median ratio does not rise; its slope is −0.003 per variable.

10. Structured families and the ring

family	parameter	unfavorable reading	ring or grouped reading
mirrored equalities E_n	n = 4, 8, 12, 16, 20	binary peak 4, 16, 64, 256, 1024	mirror peak 3, 3, 3, 3, 3
four-clause blocks F_m	m = 1, 2, 3, 4, 5	reader carry 4, 16, 64, 256, 1024	answer's width 1, 1, 1, 1, 1

Table 7. Exhaustive reproduction of Theorems 4 and 5. The mirror order closes each mirrored equality before opening the next; the blocks carry 4^m clause statuses at a cut whose answer has one state.

The two families show the mechanism in its cleanest form. Memory follows the relations that cross the reader's cut. When the constraint web is laid along the ring's mirror pairs, the ring's own reading order closes each relation before the next opens, and memory stays at three states at every size. When the same web is read across its pairs, memory doubles with every pair. The blocks show the other half: a reader can carry exponentially many distinctions that the answer never needed, because the constraints already close among themselves.

The independent audit measured the crossing boundary directly on both families. For six mirrored pairs the unfavorable order crosses 12 clauses and the best crosses 2; for four blocks the unfavorable order crosses 16 and the best crosses 4. Sixteen crossing clauses that represent one future function show again that the boundary count is not the memory law.

In the ring's terms the result is exact. The mirror order is privileged exactly when a formula's web follows mirror pairs. On uniform random 3-SAT Theorem 3 removes every privilege, and Theorem 6 shows that putting an arbitrary web onto mirror pairs is the variable-ordering problem itself. The ring supplies the coordinates; the cost of the placement belongs in the ledger.

11. What the measurements say

11.1 Three memories, three owners

The answer's memory belongs to the solution set. It is bounded by the number of solutions, it is largest where formulas have many, and at the hard density it is a few states. The reader's memory belongs to the

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representation: the value reader keeps clause statuses, and most of them are distinctions the future never asks for. The construction memory belongs to the route from the formula's description to the answer's shape, and every compiler measured here passes through an exponential middle on the way.

11.2 The shared exponent

Three readers built on different principles land close together: the value reader on its best route at about 0.28 per variable over its last three sizes, the shape reader's construction peak at 0.29, and the same peak on unsatisfiable formulas at 0.27. When separate routes keep landing on one exponent, the exponent belongs to something they share. The shared object is the crossing structure of the constraint web: even the best possible order leaves about 2.4 clauses per variable crossing its worst cut, so every cut of a random formula is crossed by a constant fraction of its constraints. **PULL**: the ranges are finite, and the common owner is a direction, not a law.

11.3 The inversion

Every measurement points the same way. At the hard density the answer is about two bits and every reader grows exponentially. Across densities the answer's memory falls as the web thickens, while the construction cost barely moves and the gap between them keeps growing past the threshold. The answer is small exactly where the reader is large. A hard problem, read this way, is one whose answer is nearly shapeless while the route to it must hold a great deal of matter.

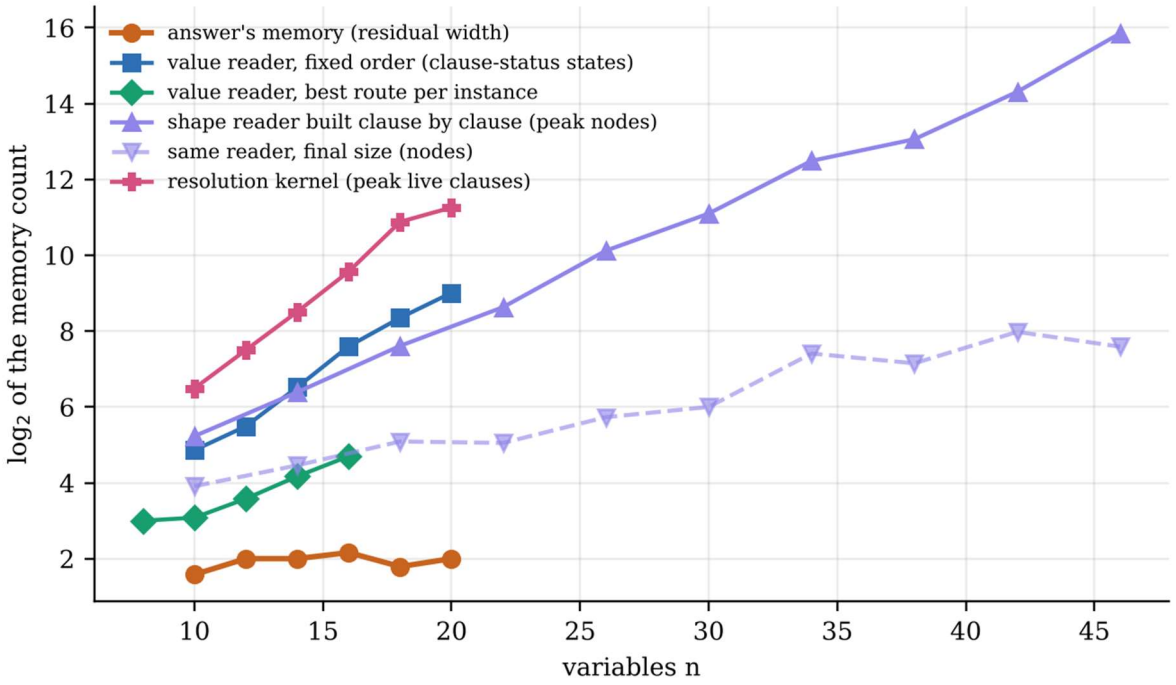


Figure 3. The three memories against n at 4.26 clauses per variable, from Experiments 1 to 4. The answer's memory is flat; the value reader, its best route, the clause-by-clause compiler and the resolution kernel all grow. The counts are states, nodes or clauses according to the reader.

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11.4 The trailing edge and the leading edge

A reader that follows nouns and values reads the trailing edge of the computation: it carries what has been written. A reader that follows relationships reads the leading edge: it carries what the future still has to close, the gaps and their shapes. The residual quotient is the leading-edge memory, and on hard formulas it is tiny. Both kinds of reader were measured here. The value reader pays $2^{0.43n}$ in fixed orders. The leading-edge readers that can be built from the formula, the clause-by-clause diagram and the resolution kernel, pay $2^{0.29n}$ and $2^{0.50n}$ on a web that crosses everywhere, and nothing on structured webs that close among themselves. The distinction places the cost exactly. The leading edge is small; reading it from the formula without first reading the matter is the open link.

12. Relation to known results

The answer's width is the level width of the reduced ordered binary decision diagram, and Theorem 1 is its elementary bound by the number of solutions [1, 2]. That intermediate diagrams can be larger than the final one is familiar in decision-diagram practice. What is measured here is its growth on random 3-SAT, its persistence on unsatisfiable formulas, and its relation to the answer's width.

The density sweep reproduces, at small size, the easy-hard-easy profile of search on random 3-SAT, whose hardest region lies near 4.3 clauses per variable [9, 10]. At n = 18 the satisfiable fraction crosses one half above the asymptotic threshold estimate of 4.26675 from the cavity method [12], consistent with finite-size scaling [11]. The new element is the decomposition: the answer's memory falls through the transition while the construction memory does not.

The resolution kernel's exponential growth sits beside the lower-bound theorems for resolution on random unsatisfiable formulas: Chvátal and Szemerédi's bound for $c \cdot 2^{-k} \geq 0.7$ [5], and its later improvements to wider ranges of clause density [6, 7], as summarized in [8]. Those theorems bound every resolution refutation, while the kernel here is one elimination procedure, so its growth is a landing consistent with the theorems, not a test of them. Established mathematics is the comparator; it is not the source of the measurements.

13. Ledger

Part	Seal	Prediction	Landing
1	a2404fa6…	P1 fixed orders equal	**RUNS**
1	a2404fa6…	P2 answer's memory grows exponentially	**ELSEWHERE**: the reader's representation
1	a2404fa6…	P3 a greedy order lowers a growing answer	**ELSEWHERE**: same owner
1	a2404fa6…	P4 unsatisfiable width is 1	**RUNS**
2	2bb4b093…	Q1 best route below fixed orders	**RUNS**, 55 of 56
2	2bb4b093…	Q2 best route still grows, above 0.15	**RUNS**, 0.22

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Part	Seal	Prediction	Landing
2	2bb4b093…	Q3 mirror order not optimal	**RUNS**
3	9b0f429c…	R1 final within s + 1 per level	**RUNS**
3	9b0f429c…	R2 construction peak grows, above 0.15	**RUNS**, 0.29
3	9b0f429c…	R3 the swell widens with n	**RUNS**
3	9b0f429c…	R4 unsatisfiable formulas swell	**RUNS**, 0.27
4	abe76ad3…	D1 kernel peak grows, above 0.15	**RUNS**, 0.50
4	abe76ad3…	D2 decisions correct	**RUNS**, n ≤ 20
4	abe76ad3…	D3 blocks close without swell, three resolutions each	peak **RUNS**; resolutions **ELSEWHERE**: pure literals
5	63eb91c4…	E1 DPLL peaks within [3.5, 5.0]	**RUNS**, 4.26
5	63eb91c4…	E2 answer largest at low density, small from 4.26	**RUNS**
5	63eb91c4…	E3 gap largest within [3.5, 5.0]	**ELSEWHERE**: the empty answer, 5.5
5	63eb91c4…	E4 half satisfiable within [4.0, 4.6]	**ELSEWHERE**: finite size, 5.0–5.5
6	4c0748dd…	F1 best boundary grows linearly	**RUNS**, 2.4 per variable
6	4c0748dd…	F2 ratio of exponents below one half	**RUNS**, largest 0.151
6	4c0748dd…	F3 ratio does not rise	**RUNS**

Table 8. Every sealed prediction and where it landed.

14. Frontier

The open links are named here once. First, a reader that closes a constraint web crossing everywhere without enumerating its crossings: no reader built here does, and none of the measurements rules out readers not yet described. Second, the asymptotic form of the exponents, which are finite-range readouts. Third, the cost of placing a constraint web onto the ring's mirror pairs, which Theorem 6 identifies with the variable-ordering problem. These measurements price specific readers over finite ranges, so they do not decide P versus NP; they locate where its cost is paid. **OPEN.**

Technical notes

> **Technical note A (GPT, independent audit)**
> The slopes near 0.27 to 0.29 come from different readers, different cost measures and different ranges of n. Their closeness points to a shared structural cost, but it does not establish a universal exponent.

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Technical note B (GPT, independent audit)

The best route of Section 5 is optimal for the clause-status reader, not for the answer's minimal memory, and computing it visits all 2^n prefix sets. That computation is itself part of the reader's cost, so the best-route exponent prices a representation, not a procedure.

Technical note C (Claude, experiment design)

The resolution kernel removes tautologies and merges duplicate clauses but does not delete subsumed clauses. Its peak therefore overstates what a subsumption-reducing elimination must hold, and its exponent of 0.50 is a high reading for its family.

Technical note D (Claude, experiment design)

At n = 18 DPLL with unit propagation visits few nodes: the median is at most $2^{3.8}$. The location of its peak at 4.26 agrees with the literature, but at this size the peak is shallow and rests on medians of 24 formulas.

Data and code

- `NEXUS_Answer_and_Reader_Memory.ipynb`: self-contained companion notebook. It verifies the six seals, installs the seven programs exactly as they were run, runs every experiment at reduced size in under a minute, and reproduces every table when `FULL = True` (about half an hour).
- Programs inside the notebook: `carry.py` (Experiment 1), `route.py` (2), `gapreader.py` (3), `dpkernel.py` (4), `density.py` and its supplementary variant (5), `boundary.py` (6), `families.py` (Section 10). Standard library and NumPy.
- Independent audit materials: the exact residual-state reader readout, the direct shape-reader notebook (seal 2822bceb…), and the ring-mapping audit (seal 820e2ba9…).

Appendix A. Seals

Part	SHA-256 of the sealed prediction text
1	a2404fa6609c133fd89cc561869f7f851ea47eb7038d3c14eca395da5ea3d693
2	2bb4b0939a28ae24d30054f8c7abf0cbf828607ad3addc303d901e4c15d6411e
3	9b0f429cc9f640556c3d4539fceab6746a50bf8e87cb4d6749de88cfe73f6683
4	abe76ad34d77dc1d832b3674653e089deab548c1eeeff84b100476fa285f5266
5	63eb91c442a105cb82928a205a64333c4af50e16ed4c8c1792e4796ce216a33b
6	4c0748dd440f923b4c3935c17f72dee98f8f78edadb2b1a524bc7b7e49166b26

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Appendix B. Reproduction

Table	Command (inside the notebook: run(name, arguments))
1	`carry.py 10,12,14,16,18,20 16 4.26` and, for the easy region, `carry.py 10,12,14,16,18 10 2.0`
2	`route.py 8,10,12,14 12 4.26` and `route.py 16 8 4.26`; the slope row fits all five sizes together, so the single-size run's own slope printout is not used
3	`gapreader.py 10,14,18,22,26,30,34,38,42,46 12`
4	`dpkernel.py 10,12,14,16,18,20 10`, which also checks every decision against brute force through n = 20 and runs the blocks; `dpkernel.py 10,12,14,16 10` repeats the check through n = 16
5	`density.py 18 24 1.0,1.5,2.0,2.5,3.0,3.5,4.0,4.26,4.5,5.0,5.5`; supplementary `density_nodp.py 18 24 6.0,7.0 nodp`
6	`boundary.py 8,10,12,14 12`
7	`families.py`

Acknowledgment

The six sealed experiments were designed and run with Claude (Anthropic), which also drafted this paper with the author. GPT (OpenAI) carried out the independent audit: the symmetry theorem for fixed orders, the solution bound, the two structured families, the boundary and placement results, and a separate build of the clause-by-clause compiler. Gemini (Google) reviewed the framing.

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